Nikhil

The next two hours

T2D Glucose Digital Twin

A synthetic subject. A two-hour forecast. An explicit uncertainty.

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A reading needs context

What may happen next?

Which context shaped the estimate?

When should the result be withheld?

Our subject is synthetic

293 active synthetic subjects

205 training / 44 validation / 44 test

Demonstration: training-split subjects only

Replay is not a live wearable feed

The forecasting path

01 Existing synthetic CGM and EHR

02 UKF physiological prior

03 PR-TFT forecast and risk

04 FastAPI versioned contract

05 Next.js validation and workspace

A prior and a learned correction

UKF prior: a simplified physiological trajectory

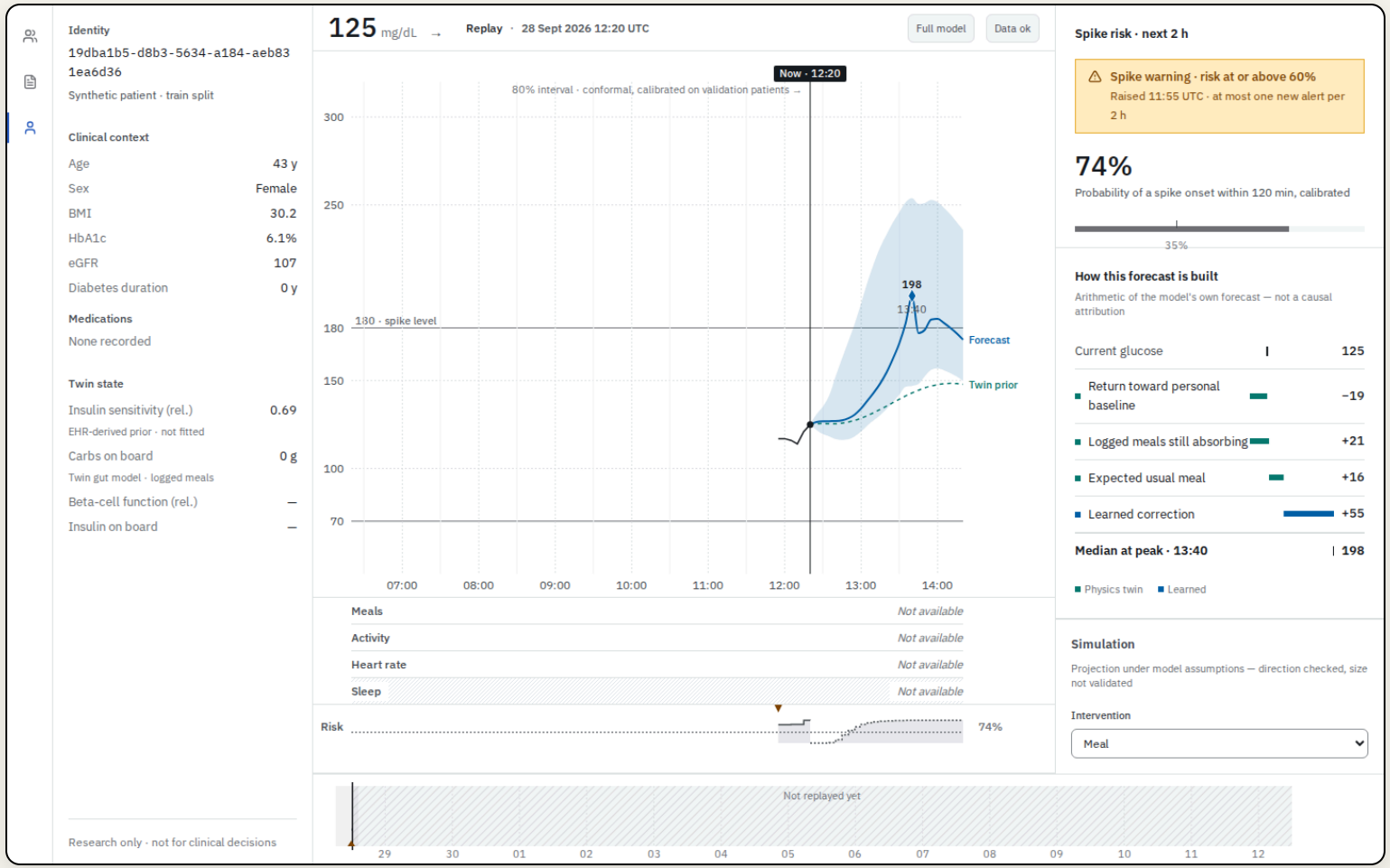
PR-TFT: learned forecasting from observable context

Checkpoint identity does not prove clinical validity

Training manifest records a dirty working tree

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Reading a forecast



120-minute horizon, five-minute steps

Median and uncertainty answer different questions

Recorded synthetic replay

Recorded screenshot: synthetic subject P0001 (training split); provenance in notes. Synthetic research only. No clinical validation claimed.

One subject, successive updates

Connected synthetic application

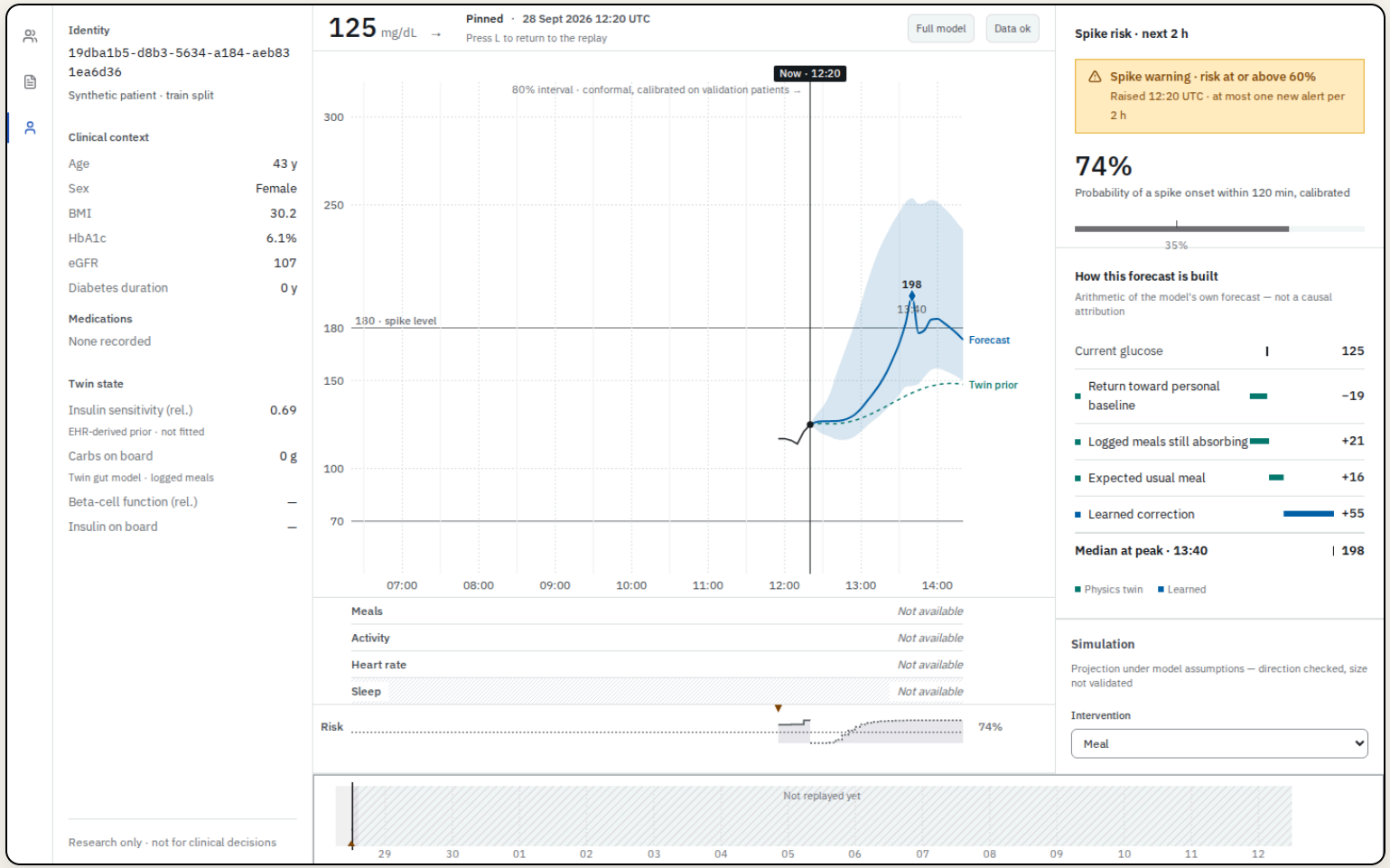
Identity / timestamp / quality / forecast

Forecast updates at recorded timestamps

Training-split subjects only; never the sealed test split

Akshaya

Holding a moment in time



A pinned view keeps the forecast anchor explicit

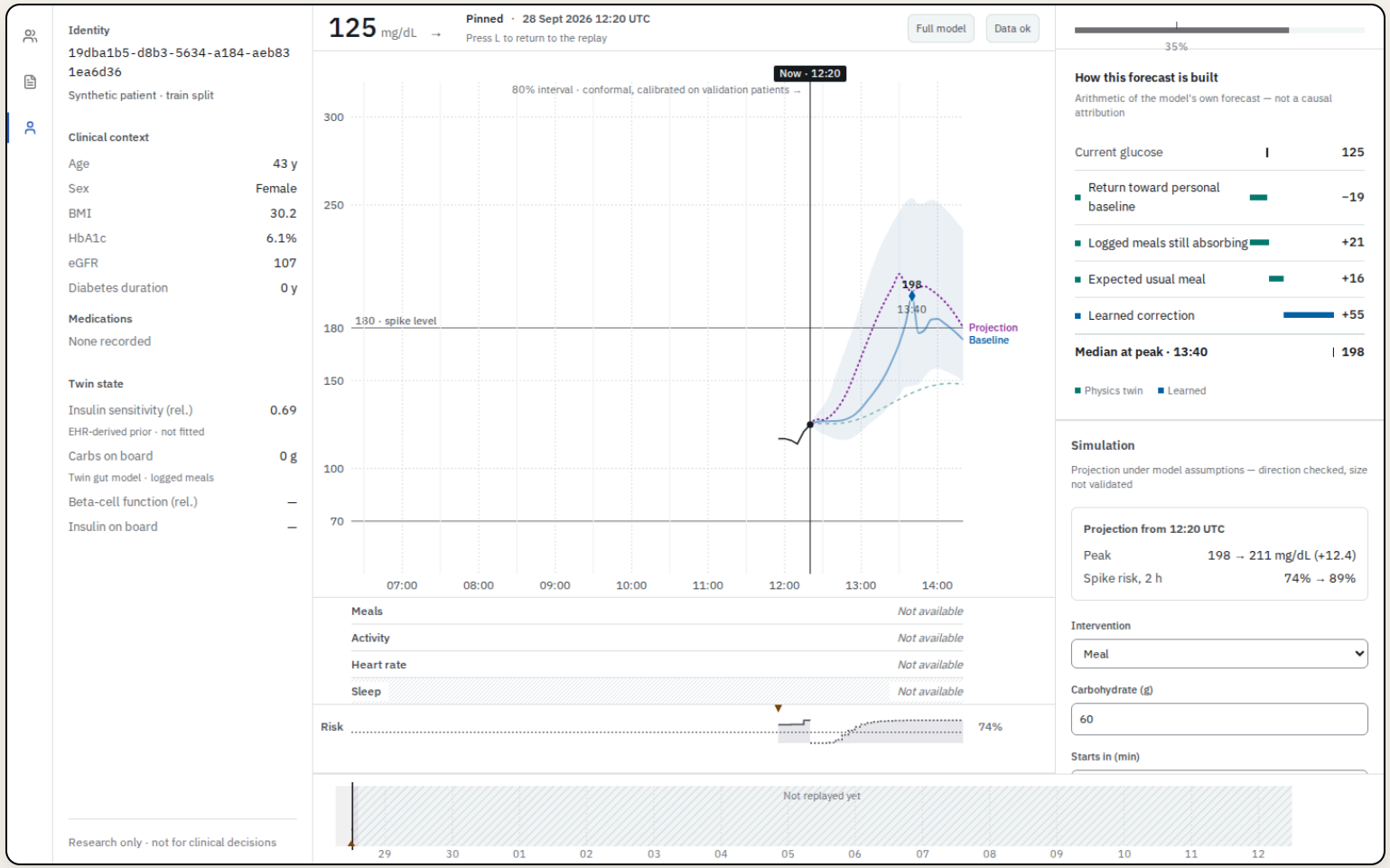
Timeline navigation uses received history

Recorded repository state, not a new execution

Recorded screenshot: synthetic subject P0001 (training split); provenance in notes. Synthetic research only. No clinical validation claimed.

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Scenario projections



Meal and activity projections exist in source

Medication is not modelled

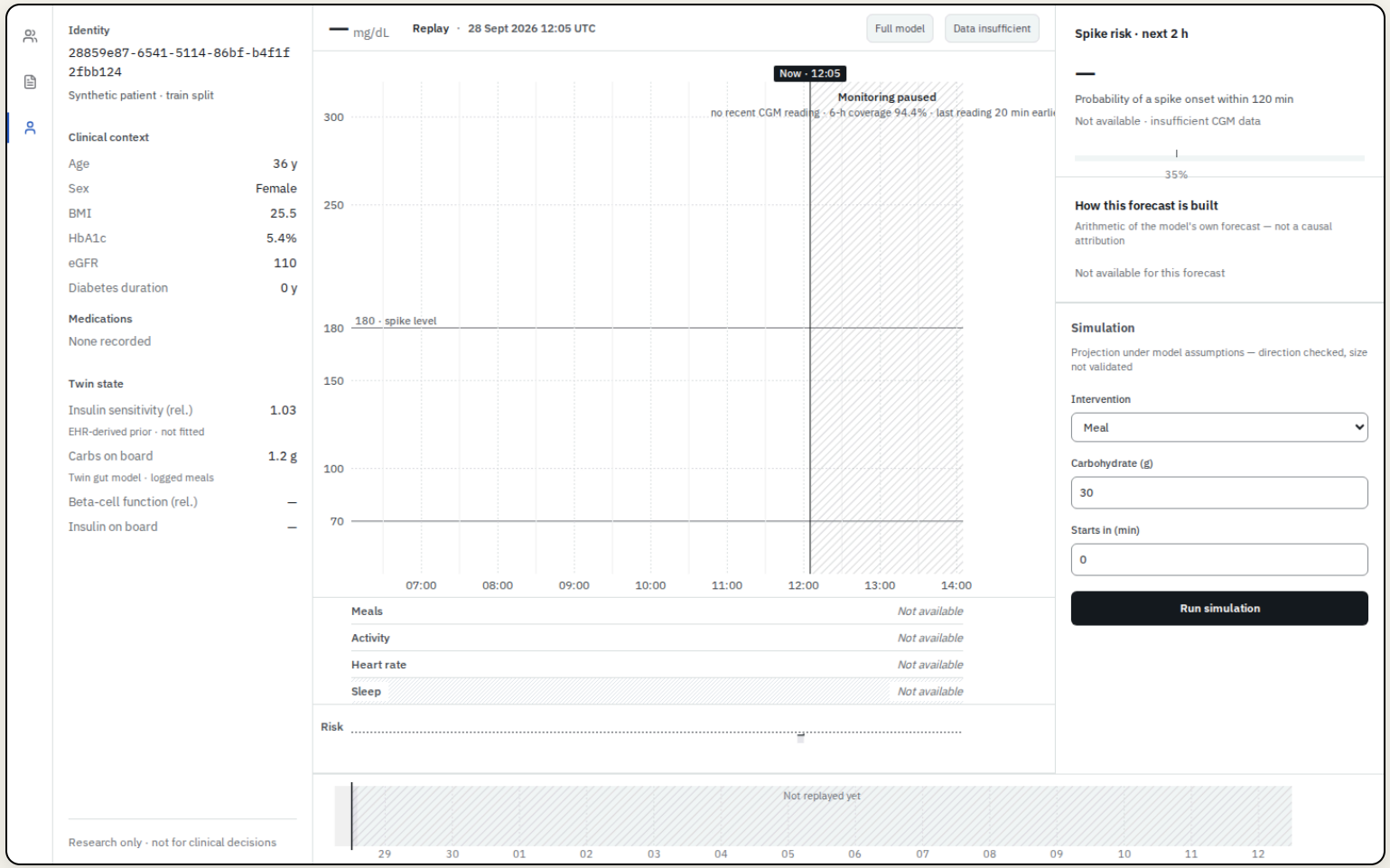
Effect size is not clinically validated

Recorded 60 g example; not treatment advice

Recorded screenshot: synthetic subject P0001 (training split); provenance in notes. Synthetic research only. No clinical validation claimed.

Akshaya

When the answer is unavailable



Insufficient history can mean no forecast

Stale, disconnected and invalid are distinct states

Reconnect and fallback need further verification

Recorded screenshot: synthetic subject P0095 (training split); provenance in notes. Synthetic research only. No clinical validation claimed.

What the recorded metrics say

Synthetic validation split (44 patients), also used for checkpoint selection and calibration: development evidence, not a held-out test

60-min RMSE: hybrid **19.53** · twin only **23.23** · persistence **36.05** mg/dL

Hybrid RMSE at 30 / 60 / 120 min: 12.36 / 19.53 / 25.03 mg/dL

Spike-risk AUROC (raw model output): 0.922

In-sample diagnostics: fitted and scored on these same validation patients

80% interval coverage 80.00% (expected by construction) · width 37.30 mg/dL · calibrated Brier 0.0703. Not independent performance evidence.

Akshaya

The realism check failed

FAIL

Gate 3 synthetic realism

Mock-real comparator

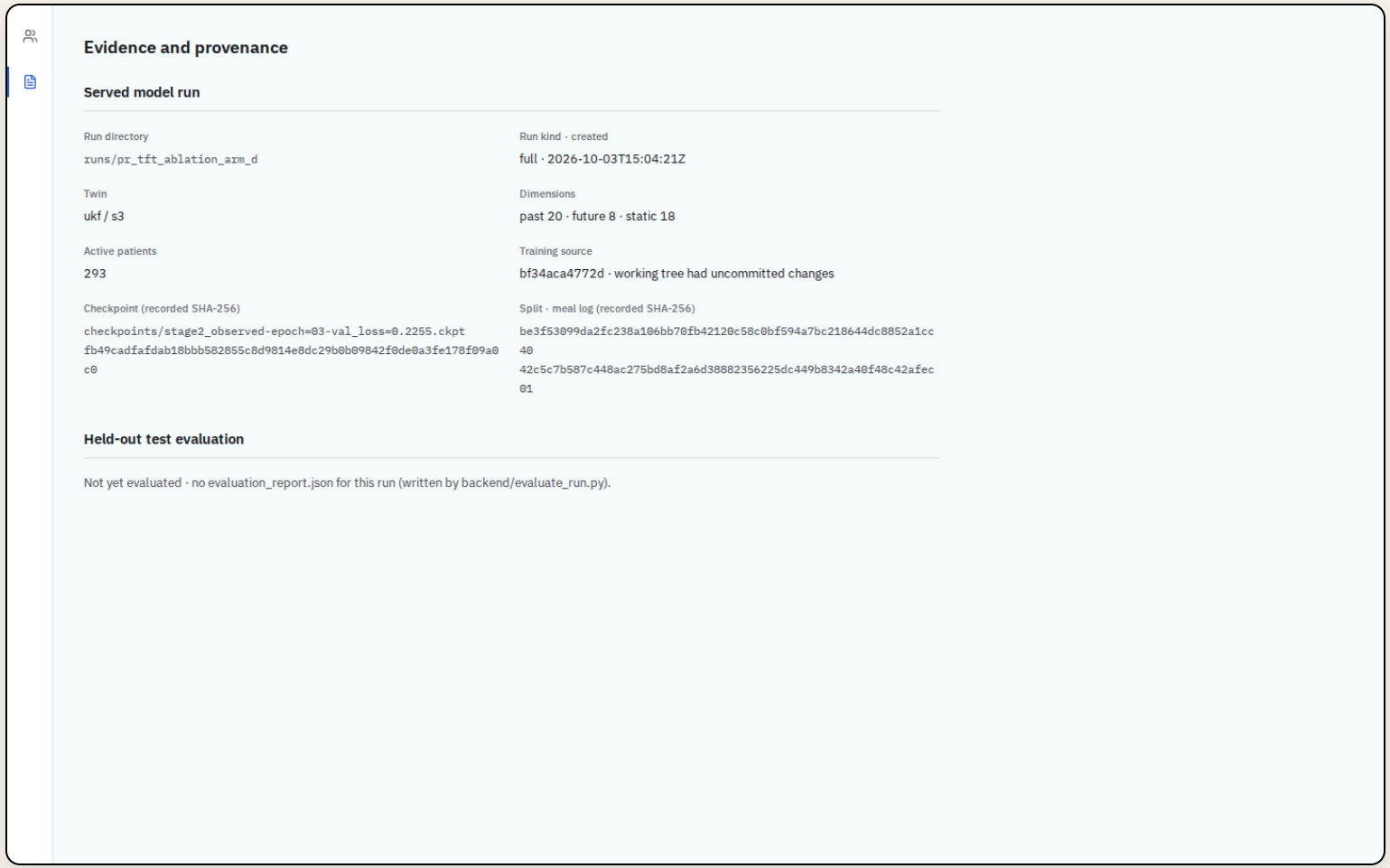
AUROC **0.932** (acceptance target ≤ 0.65)

EW-01 study protocol remains unexecuted

Clinical efficacy is unestablished. Protocol amendments remain pending co-author sign-offs; no human efficacy or cross-center transfer is claimed.

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What a reviewer can inspect



Source, contracts and manifest are inspectable

Validation metrics and held-out evaluation differ

Provenance supports review, not clinical efficacy

Recorded Evidence-page screenshot from an earlier build; provenance in notes. Synthetic research only. No clinical validation claimed.

The next two hours, with evidence

An inspectable forecasting workflow, with its evidence limits stated

Locked real-patient test (11 patients): only 6.9% of spikes warned ≥30 min ahead (95% CI 0.0–13.9; full-test point estimate, bootstrap mean 6.0%)

Two preregistered external checks (Hall 2018; AZT1D): prespecified early-warning improvement not confirmed

Candidate C2: exploratory, not established (+4.5 points, 95% CI −2.5 to +12.8)

Next: AI-READI confirmation. Protocol frozen; not yet run, data access pending

Akshaya R and Nikhil VS · **Thank you**